

Informe de experimento: llm_only_forced_structured

Generado el 2026-09-16 13:58 por benchmarks/experiment_report.py a partir de 1 report(s) del directorio llm_only_forced_structured (ruta completa en la sección 1). Prosa en español; identificadores (métodos, métricas, campos) en inglés tal como aparecen en los datos.

1. Qué se corrió

Directorio analizado:

```
/Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-f  
↪ low-agent/benchmarks/results_validation_n40/gpt-5.4/llm_only_forced_structured
```

- Reports leídos: 1 (report.rescored.json usado en 1; el resto report.json)
- Modelos: openrouter:openai/gpt-5.4
- Métodos: llm_only_forced:structured
- Casos: case14
- Requests: generadas, N=40 por caso y seed, difficulties=todas (1 report)
- Items por dificultad: plain=10, parameterized=10, multistep=10, ambiguous=10
- runs por item: 1; k (perturbación $\pm 10\% \cdot k$): 1; seeds: [0]; max_rounds: 8; temperature: 0.0; timeout_s: 90.0
- Fecha de la corrida (mtime de report.json): 2026-09-16 13:47
- Costo total: 0.6674 USD (40 items; costo medio 0.01668 USD/item)
- Items totales: 40; ok=40; con error=0; tokens totales=112674
- Solved global: 70.0% (28/40)

Errores por tipo:

(ningún item con error)

2. Prompts usados

Los prompts se reconstruyen offline con el mismo código del runner (llm.prompt_variants.build_messages sobre la red perturbada con el seed del item). Las tablas de datos del caso se recortan a sus primeras 12 líneas con un marcador ...; el número de caracteres indicado corresponde al prompt completo.

Método llm_only_forced:structured

System prompt —565 caracteres; strategy=structured, forced=True, sin herramientas

You are a power system analysis expert. You need to calculate AC power flow results based on test system data provided by the user.

Important Constraints:

- You must strictly output JSON (no extra text, no Markdown blocks).
- Do not output NaN/Infinity; all values must be finite real numbers.
- You must compute and report your best numerical estimate for every bus voltage and every branch flow. `converged` must be true and every array must be complete (one entry per bus and per branch). Do not leave arrays empty and do not state that you cannot compute.

User prompt (primer item) —5127 caracteres; case=case14, seed=1911784258, k=1, request_id=case14-ambiguous-003-s0

System base: 100 MVA. Every per-unit quantity below (r_pu, x_pu, b_pu) is defined on this base. Line and transformer impedances (r_pu, x_pu, b_pu) are already in per unit on this base, not ohms per km; do not convert them further. Branch thermal ratings are not defined for this case, so branch loading is undefined: report loading_percent as 0 for every branch.

You are a power systems expert. Based on the following IEEE case14 test system data, calculate the AC power flow results.

System Data (from pandapower network tables)

Node Data (Bus Table: net.bus)

	name	type	zone	in_service
0	1	b	1.0	True
1	2	b	1.0	True
2	3	b	1.0	True
3	4	b	1.0	True
4	5	b	1.0	True
5	6	b	1.0	True
6	7	b	1.0	True
7	8	b	1.0	True
8	9	b	1.0	True
9	10	b	1.0	True

```
10 11 b 1.0 True
... (3 líneas omitidas de 15)
```

```
### Load Data (Load Table: net.load)
```

	bus	p_mw	q_mvar	in_service
0	2	21.473798	12.567615	True
1	3	85.726702	17.290948	True
2	4	45.568126	-3.717902	True
3	5	6.887322	1.449962	True
4	6	11.740497	7.861940	True
5	9	32.229269	18.135792	True
6	10	9.714580	6.260507	True
7	11	3.424253	1.761044	True
8	12	6.539657	1.715320	True
9	13	14.757610	6.340306	True
10	14	15.783969	5.296634	True

```
### Generator Data (Generator Table: net.gen)
```

	bus	p_mw	vm_pu	min_p_mw	max_p_mw	min_q_mvar	max_q_mvar	in_service
0	2	41.63323	1.045	0.0	140.0	-40.0	50.0	True
1	3	0.00000	1.010	0.0	100.0	0.0	40.0	True
2	6	0.00000	1.070	0.0	100.0	-6.0	24.0	True
3	8	0.00000	1.090	0.0	100.0	-6.0	24.0	True

```
### Balanced Node/External Grid (External Grid Table: net.ext_grid)
```

	bus	vm_pu	va_degree	in_service
0	1	1.06	0.0	True

```
### Line Data (Line Table: net.line)
```

	from_bus	to_bus	r_pu	x_pu	b_pu	parallel	in_service
0	1	2	0.01938	0.05917	0.0528	1	True
1	1	5	0.05403	0.22304	0.0492	1	True
2	2	3	0.04699	0.19797	0.0438	1	True
3	2	4	0.05811	0.17632	0.0340	1	True
4	2	5	0.05695	0.17388	0.0346	1	True
5	3	4	0.06701	0.17103	0.0128	1	True
6	4	5	0.01335	0.04211	0.0000	1	True
7	6	11	0.09498	0.19890	0.0000	1	True
8	6	12	0.12291	0.25581	0.0000	1	True
9	6	13	0.06615	0.13027	0.0000	1	True
10	9	10	0.03181	0.08450	0.0000	1	True

... (4 líneas omitidas de 16)

```
### Transformer Data (Transformer Table: net.trafo)
```

	hv_bus	lv_bus	r_pu	x_pu	shift_degree	tap_side	tap_neutral	tap_min	tap_max	tap_step_percent	in_service
0	4	7	0.0	0.20912	0.0	hv	0.0	NaN	NaN	2.2	True
1	4	9	0.0	0.55618	0.0	hv	0.0	NaN	NaN	3.1	True
2	5	6	0.0	0.25202	0.0	hv	0.0	NaN	NaN	6.8	True
3	7	8	0.0	0.17615	0.0	None	NaN	NaN	NaN	NaN	True
4	7	9	0.0	0.11001	0.0	None	NaN	NaN	NaN	NaN	True

```
## Task
```

```
Load case14 and disconnect the line between bus ten and bus eleven.
```

```
## Output Requirements
```

```
Please calculate and return the following results (JSON format):
```

- 1) Voltage magnitude (p.u.) and phase angle (degrees) for each node
- 2) Active power (MW) and load factor (%) at the beginning of each branch
 - For lines: `line\_id` uses the row index of `net.line`
 - For transformers: `line\_id` uses the row index of 100000 + `net.trafo`
- 3) Total generation (MW), total load (MW), total losses (MW)
- 4) Are there any voltage overruns (<0.95 or >1.05 p.u.) or line overruns (>100%)?

```
Please strictly follow the following JSON schema output (all fields are complete and of correct type):
```

```
{
  "converged": true/false,

  "bus_voltages": [{"bus_id": 1, "vm_pu": 1.0, "va_deg": 0.0}, ...],

  "line_flows": [{"line_id": 0, "p_from_mw": 0.0, "loading_percent": 0.0}, ...],

  "total_generation_mw": 0.0,

  "total_load_mw": 0.0,
  "total_loss_mw": 0.0
}
```

[Important] bus_id must use the `name` column of net.bus (the usual IEEE/MATPOWER 1..N numbering), not the row index of net.bus.

3. Resultados agregados

Columnas del protocolo (medias por método $\times$ caso, del `scoreboard_per_case`), en los 4 grupos de `metricas_pflagent_definicion`
utilidad de tarea Form. (formulación exacta de las tool calls), V_MAE (p.u.) y F_MAE (MW) sobre los items
con resultado, Solved (respondió correctamente de extremo a extremo); **corrección solver-grounded** Solver status
(convergencia reportada = convergencia real), B_mean (residual medio de KCL de los flujos reportados, MW); **fidelidad**
Faith. (números de la respuesta trazables a salidas de tools) y SFR (fallos declarados con seguridad); **costo** Calls (tool
calls medias) y Tok. (tokens medios). Fuera de los 4 grupos, solo en este reporte (no en las tablas del paper): Claim
(éxito reclamado sobre un fallo), Abst. (abstención en items resolubles) y \$ (costo total USD). Porcentajes en %.

Por método $\times$ caso

Method	Case	N	Form.V	MAEF	MAE	Solved	Solver status	B_meanFaith.	SFR	Calls	Tok.	ClaimAbst.	V pass	\$	
llm_only_forced:structured	case14	40	–	2.74e-03	5.45	70.0 (28/40)	100.0 (40/40)	9.86e+00	0.6	–	n/a	2817	0.0 (0/40)	0.0 (0/40)	0.6674

Formulación exacta por dificultad (items con etapa de tools; n/a = sin tools)

Method	plain	parameterized	multistep	ambiguous
llm_only_forced:structured	n/a	n/a	n/a	n/a

Solved por dificultad (% , items resueltos / items)

Method	plain	parameterized	multistep	ambiguous
llm_only_forced:structured	90.0 (9/10)	80.0 (8/10)	40.0 (4/10)	70.0 (7/10)

Tipos de error de formulación (conteo de items)

(ningún item con `formulation_error_type`)

Errores de ejecución por método (conteo de items con error)

(ningún item con error)

4. Trazas de ejemplo

Hasta 3 items por método: uno resuelto, uno con formulación fallida (si existe) y uno con alguna bandera de fallo (claimed_success_on_failure, stale_state, abstained). Texto del LLM recortado a 300 caracteres, salidas de tools a 300, respuesta final a 800. Se lee la traza completa de **traces/** cuando existe.

Método llm_only_forced:structured

Ejemplo: resuelto (solved=True)

```
request (case14-ambiguous-011-s0, case14, difficulty=ambiguous, seed=1971893210): Load case14 and set the active load at bus 6 to 14,700 kw.
intended_calls: load_case({"case_name": "case14"}) -> modify_load({"bus_id": 6, "p_mw": 14.7})
executed_calls: (sin etapa de tools)
fuente de la traza: archivo de traza completo
(sin rondas de herramientas: una sola llamada al LLM)
--- Respuesta final ---
{"converged":true,"bus_voltages":[{"bus_id":1,"vm_pu":1.06,"va_deg":0.0}, {"bus_id":2,"vm_pu":1.045,"va_deg":-4.98}, {"bus_id":3,"vm_pu":1.01
,"va_deg":-12.72}, {"bus_id":4,"vm_pu":1.017,"va_deg":-10.33}, {"bus_id":5,"vm_pu":1.019,"va_deg":-8.78}, {"bus_id":6,"vm_pu":1.07,"va_deg
-> "-14.22}, {"bus_id":7,"vm_pu":1.062,"va_deg":-13.37}, {"bus_id":8,"vm_pu":1.09,"va_deg":-13.37}, {"bus_id":9,"vm_pu":1.056,"va_deg":-14.9
-> 4}, {"bus_id":10,"vm_pu":1.051,"va_deg":-15.1}, {"bus_id":11,"vm_pu":1.057,"va_deg":-14.79}, {"bus_id":12,"vm_pu":1.055,"va_deg":-15.08}, {
-> "bus_id":13,"vm_pu":1.05,"va_deg":-15.16}, {"bus_id":14,"vm_pu":1.036,"va_deg":-16.04}], "line_flows": [{"line_id":0, "p_from_mw":156.88, "l
-> oading_percent":0.0}, {"line_id":1, "p_from_mw":75.51, "loading_percent":0.0}, {"line_id":2, "p_from_mw":73.24, "loading_perce
-> n ... [1020 chars
-> más]
--- Scoring ---
formulation=m/a (exact=None); V_MAE=4.27e-04; F_MAE=2.25; faithful=0.0 %; solved=True (numeric_ok); flags=-; error=-; tokens=2805;
-> cost usd=0.0166; wall=6.6 s
```

Ejemplo: no resuelto

```
request (case14-ambiguous-003-s0, case14, difficulty=ambiguous, seed=1911784258): Load case14 and disconnect the line between bus ten and
↳ bus eleven.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
executed_calls: (sin etapa de tools)
fuente de la traza: archivo de traza completo
(sin rondas de herramientas: una sola llamada al LLM)
--- Respuesta final ---
{"converged":true,"bus_voltages":[{"bus_id":1,"vm_pu":1.06,"va_deg":0.0}, {"bus_id":2,"vm_pu":1.045,"va_deg":-4.98}, {"bus_id":3,"vm_pu":1.01
↳ ,"va_deg":-12.72}, {"bus_id":4,"vm_pu":1.017,"va_deg":-10.33}, {"bus_id":5,"vm_pu":1.019,"va_deg":-8.78}, {"bus_id":6,"vm_pu":1.07,"va_deg
↳ ":-14.22}, {"bus_id":7,"vm_pu":1.062,"va_deg":-13.37}, {"bus_id":8,"vm_pu":1.09,"va_deg":-13.36}, {"bus_id":9,"vm_pu":1.056,"va_deg":-14.9
↳ 4}, {"bus_id":10,"vm_pu":1.051,"va_deg":-15.1}, {"bus_id":11,"vm_pu":1.082,"va_deg":-14.79}, {"bus_id":12,"vm_pu":1.055,"va_deg":-15.08}, {
↳ "bus_id":13,"vm_pu":1.05,"va_deg":-15.16}, {"bus_id":14,"vm_pu":1.036,"va_deg":-16.04}], "line_flows": [{"line_id":0, "p_from_mw":156.88, "l
↳ oading_percent":0.0}, {"line_id":1, "p_from_mw":75.51, "loading_percent":0.0}, {"line_id":2, "p_from_mw":73.24, "loading_perce
↳ n ...[1023 chars
↳ más]
--- Scoring ---
formulation=n/a (exact=None); V_MAE=3.90e-03; F_MAE=2.12; faithful=0.0 %; solved=False (voltage_error); flags=-; error=-; tokens=2800;
↳ cost_usd=0.0166; wall=7.2 s
```

Ejemplo: primer item

```
request (case14-ambiguous-007-s0, case14, difficulty=ambiguous, seed=2073320062): Load the 14-bus test case (case14) and disconnect the
↳ line between bus 6 and bus 13 (1-based).
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 6, "to_bus": 13})
executed_calls: (sin etapa de tools)
fuente de la traza: archivo de traza completo
(sin rondas de herramientas: una sola llamada al LLM)
--- Respuesta final ---
{"converged":true,"bus_voltages":[{"bus_id":1,"vm_pu":1.06,"va_deg":0.0}, {"bus_id":2,"vm_pu":1.045,"va_deg":-4.98}, {"bus_id":3,"vm_pu":1.01
↳ ,"va_deg":-12.72}, {"bus_id":4,"vm_pu":1.017,"va_deg":-10.33}, {"bus_id":5,"vm_pu":1.019,"va_deg":-8.78}, {"bus_id":6,"vm_pu":1.07,"va_deg
↳ ":-14.22}, {"bus_id":7,"vm_pu":1.062,"va_deg":-13.37}, {"bus_id":8,"vm_pu":1.09,"va_deg":-13.36}, {"bus_id":9,"vm_pu":1.056,"va_deg":-14.9
↳ 4}, {"bus_id":10,"vm_pu":1.051,"va_deg":-15.1}, {"bus_id":11,"vm_pu":1.057,"va_deg":-14.79}, {"bus_id":12,"vm_pu":1.055,"va_deg":-15.08}, {
↳ "bus_id":13,"vm_pu":1.05,"va_deg":-15.16}, {"bus_id":14,"vm_pu":1.036,"va_deg":-16.04}], "line_flows": [{"line_id":0, "p_from_mw":156.88, "l
↳ oading_percent":0.0}, {"line_id":1, "p_from_mw":75.52, "loading_percent":0.0}, {"line_id":2, "p_from_mw":73.24, "loading_perce
↳ n ...[1022 chars
↳ más]
--- Scoring ---
formulation=n/a (exact=None); V_MAE=8.98e-03; F_MAE=9.87; faithful=0.0 %; solved=False (voltage_error); flags=-; error=-; tokens=2814;
↳ cost_usd=0.0167; wall=7.0 s
```

5. Lecturas

Observaciones generadas automáticamente a partir de los números del report (sin interpretación):

- LLM-only: V_MAE entre 2.74e-03 (llm_only_forced:structured/case14) y 2.74e-03 (llm_only_forced:structured/case14) p.u. sobre los items que devolvieron números.
- 0.0 % (0/40) de abstención en LLM-only (respuesta con converged=false o arrays vacíos).
- En LLM-only, 0 items no resueltos por convergence_mismatch y 12 por voltage_error.
- Ningún item terminó con error.
- Solved de llm_only_forced:structured: 70.0 % (28/40).
- Ningún item reclamó éxito sobre un fallo (claimed_success_on_failure=0).
- Trazabilidad de números (Faith.) media en LLM-only: 1.6 % (sin tools, todo número es no trazable salvo los copiados de la request).
- Costo total 0.6674 USD; el método más caro fue llm_only_forced:structured con 0.6674 USD (100 %).

6. Archivos

Rutas absolutas bajo llm_only_forced_structured. report.rescored.json (cuando existe) es el archivo leído para las métricas.

```
report: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2
↳ /power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/llm_only_forced_structured/report.json [leído:
↳ report.rescored.json]
scoreboard.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGr
↳ ids-Code-v2/power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/llm_only_forced_structured/scoreboard.json
scoreboard.csv: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrid
↳ ds-Code-v2/power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/llm_only_forced_structured/scoreboard.csv
scoreboard.md: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrid
↳ s-Code-v2/power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/llm_only_forced_structured/scoreboard.md
scoreboard_per_case.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-Fo
↳ r-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/llm_only_forced_structured/scoreboard_per_case.json
scoreboard.rescored.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-Fo
↳ r-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/llm_only_forced_structured/scoreboard.rescored.json
scoreboard_per_case.rescored.json:
↳ /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/p
↳ ower-flow-agent/benchmarks/results_validation_n40/gpt-5.4/llm_only_forced_structured/scoreboard_per_case.rescored.json
```

```
traces (40 archivos): /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-Sm
↳ artGrids-Code-v2/power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/llm_only_forced_structured/traces
```